

SECTOR: ICT

SUB-SECTOR: IT

ACADEMIC YEAR:2025-2026

FYP PROPOSAL GUIDELINE

PART I: INTRODUCTION

The Final Year Project (FYP) is an essential component of undergraduate education, designed to challenge students by involving them in the process of solving real-world problems. Drawing from Articles 202, 216 and others of the Student Handbook 2023, this guideline will help students understand the purpose of the FYP, the benefits it offers to both students and the institution, and the key components that should be included in the FYP proposal.

The main objective of the FYP is to enable students to identify, investigate, and tackle practical issues within their field of study. By choosing a real-life problem to address, students can apply the theoretical knowledge they have acquired throughout their course in a hands-on way. The FYP is not just an academic requirement but a platform to showcase creativity, critical thinking, and problem-solving skills. Through their projects, students are encouraged to make innovative contributions that add value to their discipline, which in turn prepares them for their future careers.

For students, the FYP is an opportunity to deepen their expertise and gain practical experience in their field. By engaging in research and application, they develop skills that are essential for the workplace, such as project management, analysis, and technical proficiency. The project also helps students build a portfolio of work that can be showcased to potential employers.

On the institutional side, the FYP reflects the college's commitment to quality education that prepares students for industry demands. Successful projects can enhance the college's reputation and strengthen its connections with industry partners, as they demonstrate that students are equipped to address real-world challenges. This alignment with industry standards also opens doors for future collaborations, internships, and job opportunities for students.

PART II: FYP REGULATIONS AND STUDENTS ELIGIBILITY

1. FYP regulations

The Final Year Project (FYP) is a mandatory component, governed by formal rules to ensure that students meet high standards in their research and presentation. These regulations cover the following areas:

1.1 Submission Deadlines

Students must submit their project proposals, reports, and final project documents by dates specified in the academic calendar. Late submissions will incur a 'zero' mark without the possibility of extension unless approved by the College Academic Board (Article 217). Final corrected reports must be submitted not later than two weeks after the defense (Article 215).

1.2 Required Documentation

Each candidate must submit two hard copies of the project report and one soft copy on a CD (*CD is needed after corrections and after final defense*). One hard copy will be stored in the library, and the other may be retained by the student if desired (Article 215).

1.3 Project Scope Limitations

The length of the project report should be between 4,000 to 6,000 words, excluding ancillary material like tables and appendices (Article 205) [*This must be respected on the final defense not proposal. The proposal can be 3,000 words approximately*]. The project must be an industry-related applied research project (Article 216) and must contribute to knowledge or practice in the student's field.

1.4 Plagiarism standards

Final Year Projects must strictly adhere to the institution's plagiarism policy, as outlined in the RP student handbook. Any content submitted must be original, with proper citations for all sources. Specific penalties for plagiarism in Final Year Projects include:

- **Major Plagiarism (over 49% plagiarized):** Results in serious consequences, including rejection of the project, overall module failure, suspension for up to a year, or even discontinuation from studies. In severe cases, it may lead to revocation of previously awarded degrees or credentials (Article 155).
- **Moderate Plagiarism (31-49% plagiarized):** Leads to a zero on the plagiarized work and a formal written warning (Article 155).
- **Minor Plagiarism (0-30% plagiarized):** No penalties are imposed, but students are encouraged to revise to meet originality standards (Article 155).

Additionally, if draft work is submitted solely for feedback rather than grading, any detected plagiarism is not considered an offense. Supervisors will advise students on maintaining academic integrity and may inform examiners as necessary (Article 156).

1.5 Evaluation Criteria

Projects will be assessed based on a structured format approved by the department. The assessment will include the proposal, written report, and oral defense (Articles 216, 212). Projects are to be evaluated by a panel, consisting of three trainers and the supervisor, who will judge the student's presentation and project quality (Article 216).

2. Students' eligibility

To qualify for the Final Year Project, students must meet the following criteria:

2.1 Academic Standing

Students should be in good academic standing, with no outstanding academic probation or disciplinary actions.

2.2 Prerequisite Courses and Credits

Candidates must have completed all prerequisite courses related to their major, as determined by their department. They should have accrued the minimum required credits for final-year status, confirming their readiness for an industry-related project.

3. User stories / special cases

To accommodate diverse student needs, the FYP guidelines address several special cases:

- **Students without regular supervisor access:** For students who cannot meet with their supervisors regularly, departments may provide alternative support, such as virtual check-ins or a designated secondary advisor.
- **Late submission of project topic:** Students who are delayed in submitting their topics for validation may face limitations in the choice of project and a delayed start date, impacting the project timeline.
- **Students with disabilities:** The department will offer appropriate support, such as extended time for proposal preparation or adaptive technology during presentations, ensuring these students can complete their projects successfully.
- **Group projects with varying roles:** For projects with multiple students, each member's specific role and contribution must be documented in the proposal. The assessment will be based on individual contributions as well as the overall project outcome (Article 220).
- **Interdisciplinary Projects:** When projects span multiple disciplines, students may work with co-supervisors from each field, ensuring balanced guidance and assessment across areas.

4. Make-up procedures for missed proposal presentations

Students who miss their scheduled proposal presentations have a structured procedure for rescheduling:

- **Documentation requirements:** Students must provide documentation justifying their absence, such as medical certificates or other formal proof, to avoid penalties.
- **Rescheduling policies:** Students may reschedule their proposal defense once, with the new date set no later than two weeks from the initial assessment date (Article 214).

PART III: TOPIC SELECTION AND VALIDATION

1. Topic selection

When selecting a topic for your Final Year Project (FYP) in IT, consider focusing on areas such as:

- IoT with control web apps,
- Web of Things (WoT),
- AI and machine learning,
- Networking systems,
- Augmented and virtual reality (AR/VR),
- Human-computer interaction,
- Web-based app development,
- Mobile Application Development
- Blockchain technology.

Potential projects could include developing IoT systems controlled through web interfaces, creating machine learning models for data analytics, optimizing network security and performance, or building AR/VR applications for immersive training or simulations. Additionally, you could explore integrating these technologies, such as combining blockchain with IoT for secure data sharing or using AI for intelligent network management.

2. Topic validation

To get a chosen topic approved, students must submit three (3) topic proposals within the first two weeks of the semester, using a Topic Proposal Form that outlines the topic's relevance and expected outcomes. After submission, students meet with their assigned academic supervisor for feedback. The supervisor will either approve a topic or suggest revisions based on its quality and productivity. Students must revise their topics as needed and resubmit them by the end of the third(3) week.

Students must be sure that their topics are feasible, align with the department's focus, and fit their interests and skills. If it's a group, a project must be advanced and worthy to be done by more than one student. The joint of students (group) must be approved by the department after reviewing the heaviness of the project.

3. Criteria for topic evaluation

When assessing the suitability of proposed topics, the following criteria are used:

- **Originality:** The topic must be unique, offering fresh ideas or approaches not previously explored.
- **Relevance to the IT field:** The topic must be an IT solution, even if it addresses challenges in other domains (e.g., healthcare, education, finance).
- **Potential for research:** The topic should provide opportunities for in-depth investigation, innovation, and contribution to the IT field.
- **Feasibility:** The topic must be achievable within the given timeframe and with the available resources (e.g., software, hardware, data).
- **Alignment with departmental goals:** The topic should align with the department's focus areas and academic standards.
- **Innovation and impact:** The topic should demonstrate the potential for technological advancement or real-world impact.
- **Practical application:** The topic should have practical relevance, with clear applicability to real-world IT solutions, even if applied in another domain.

PART IV: FORMATTING STANDARDS

1. Page Layout and General Formatting

For consistent formatting and a professional look of all report pages excluding the cover page, the following specifications should be followed:

1. Margin sizes:

- **Top:** 1 inch (2.54 cm)
- **Bottom:** 1 inch (2.54 cm)
- **Left:** 1 inch (2.54 cm)
- **Right:** 1 inch (2.54 cm)

2. Page orientation:

- **Portrait** orientation is required for all pages.

3. Footer requirements:

- **Footer:** Should include page numbers centered.

4. Font type and size:

- **Font type:** Times New Roman.
- **Font Size:**
 - **Main text:** 12-point size.
 - **Heading 1:** 14-point, bold.
 - **Other headings:** 12-point size, bold.
 - **Figure and table captions:** 12-point size, italic.

5. Line spacing:

- **Main text:** 1.5 line spacing for readability.
- **Headings:** Single line spacing, with a space before and after each heading.
- **Abstract:** Single line spacing.

6. Paragraph alignment:

- **Main text:** Justify-aligned.
- **Paragraph Indentation:** Indentation of 0.5 inch (1.27cm) is required only for the lists of any heading(e.g.: Objectives, research questions, etc).

7. Colors:

- Text color for all is **Black**.

2. Tables, Figures, and Table of Contents

2.1. Labeling and Formatting Tables and Figures

a. Placement

- Tables and figures should be placed as close as possible to the relevant text in the document, preferably within the same section where they are referenced.
- Avoid placing tables and figures on separate pages unless absolutely necessary and if a table has taken more than one page, ensure it's titled on both pages.

b. Numbering

- **Tables:** Numbered consecutively throughout the document (e.g., Table 1, Table 2, etc.). Tables should be numbered separately from figures.
- **Figures:** Numbered consecutively as well (e.g., Figure 1, Figure 2, etc.).
- Use the format **Table X** and **Figure X** for labeling, where X is the number.

c. Captioning

- **Tables:** The caption should be placed above the table and should briefly describe its contents (e.g., "Table 1: list of equipment tools").
- **Figures:** The caption should be placed below the figure and should describe what the figure represents (e.g., "Figure 1: Entity Relationship Diagram").
- Both captions should be concise but descriptive enough to allow the reader to understand the content without referring to the main text.

d. Font and Formatting for Tables and Figures

- Use the same font style and size for table captions and figure captions as the main text (usually **12-point font**).
- Ensure table text is legible and aligned, with clear borders or spacing for easy reading.
- For figures, ensure they are clear, with high resolution, and properly sized to fit the page.

2.2. Table of contents

1. Use Heading Styles

- Apply built-in **heading styles** (e.g., Heading 1, Heading 2, etc.) to your section and subsection titles throughout the document. These styles are essential for the automated TOC generation.
- **Heading 1:** Main section titles (e.g., Chapter 1, Chapter 2).
- **Heading 2:** Subsections (e.g., 1.1, 1.2).
- **Heading 3:** Sub-subsections(e.g., 1.1.1, 1.1.2).
- Add sub-subsections if encountered in the report.

2. Inserting the Table of Contents

- In most word processors (e.g., Microsoft Word), use the **References** menu to add a **Table of Contents**. Choose the option for **automatic TOC** based on the headings applied.
- The TOC will automatically update based on the headings in your document, providing clear navigation.

3. References and Citations Formatting

References must be automatically added using the **References** menu feature in the word processor. Both citations and the reference list must respect APA style, ensuring consistency and accuracy. The following are additional guidelines:

3.1. In-Text citations

- **One author:** (Author, Year), e.g., (Smith, 2020).
- **Two authors:** (Author1 & Author2, Year), e.g., (Smith & Johnson, 2020).
- **Three or more authors:** (Author1 et al., Year), e.g., (Smith et al., 2020).

3.2. Reference Format

Books

- Author, A.A. (Year). *Title of the book*. Publisher.

Example: Smith, J. (2020). *Understanding data structures*. Academic Press.

Journal Articles

- Author, A.A. (Year). Title of the article. *Title of the Journal*, Volume(Issue), page range.
<https://doi.org/xx.xxx>

Example: Smith, J., & Johnson, R. (2020). Analyzing big data trends. *Journal of Data Science*, 15(3), 45-59. <https://doi.org/10.1234/jds.2020.00123>

Websites:

- Author, A.A. (Year, Month Day). Title of the web page. *Website Name*. URL

Example: Smith, J. (2020, June 15). Exploring new trends in AI. *AI Innovations*.
<https://www.aiinnovations.com/articles/trends>

PART V: CONTENT REQUIREMENTS

1. Cover page

Cover page must contain the following contents(Article 211):

- RP/College logo
- Project title [*Center-aligned, bold*]. Put vertical space between Project title and student name.
- Student name [*Center-aligned, bold, italic*]. Don't put empty space between Student name and reg. number.
- Student registration number [*Center-aligned, bold, italic*]. Put vertical space between Student reg. Number and Supervisor name

- Supervisor name [*Center-aligned, bold*]. Put a vertical space between supervisor name and academic year
- Academic year [*Center-aligned*].

Cover page must NOT have a page number

2. Preliminary page

Preliminary pages are numbered with romans in small letters(i, ii, iii, iv, ...)

DECLARATION

A statement by the student confirming that the project is original and their own work.

Example:

I ***Student_Name***, a student at RP Karongi College hereby declares that I carried out the work reported in this report in the Department of Information Communication and Technology at RP Karongi College under the supervision of ***Supervisor_Name*** I solemnly declare that to the best of my knowledge, no part of this report has been submitted anywhere in a previous application for award of an academic qualification.

Student_names

Reg: Student_registration_number

Date / /

Signature.....

BONAFIDE CERTIFICATE

A supervisor certification that the student completed the project per academic standards.

Example:

This is to certify that the project titled ***Project_title*** carried out by ***Student_Name_and_reg_number*** has been read, checked and approved for meeting part of the requirements and regulations governing the award of the Advanced Diploma in Information and communication technology at RP Karongi College.

Supervisor_Name

Signature.....

Date...../...../.....

DEDICATION

A brief note dedicating the work to someone meaningful, like family or mentors

ACKNOWLEDGEMENTS

Thanks to individuals, such as supervisors or peers, who supported or contributed to the project.

TABLE OF CONTENTS

An outline of the document structure, listing sections and their page numbers for easy navigation. *Built-in(Automatic Table 1) format should be respected.*

LIST OF FIGURES

A numbered list of all figures with their corresponding page numbers, enhancing document organization. *All figure captions must be placed directly below the figures.*

LIST OF TABLES

A similar list for tables, helping readers locate specific data tables quickly. *All table captions must be placed directly below the tables.*

ACRONYMS AND ABBREVIATION

Definitions of shortened terms used throughout the report for clarity.

ABSTRACT

The abstract should be a single, well-structured page and include the following:

- Purpose and objectives of the study
- Significance of the study
- Methodology applied
- Key findings
- Recommendations

The line spacing should be set to 1.

3. Main document

Starting from chapter 1, all pages must be numbered with numbers (1,2,3,4,...)

CHAPTER 1: GENERAL INTRODUCTION

1.1 Background of the project

This section provides a comprehensive overview of the project topic, covering its origins, importance, and relevant historical or theoretical context. It should detail existing knowledge or developments related to the project and highlight gaps that the project intends to address. *(Should be at least one page with references.)*

1.2 Statement of the problem

Here, the central issue or challenge that the project seeks to resolve is clearly stated. This section should explain why the problem is significant, outline its effects, and briefly suggest the need for a solution. It sets the stage for the project objectives and aims. (*Proof of existing or the problem as reference is required*)

1.3 Project Objectives

1.3.1 General objective

A brief, showing the main objective of the project.

1.3.1 specific objectives

These are detailed, measurable goals that break down the general objective into smaller tasks or questions the project will address.

- The objectives should be listed in bullet points, such as (i), (ii), (iii).
- They must follow the SMART criteria (Specific, Measurable, Achievable, Relevant, and Time-bound).
- Variables and their relationships should be clearly defined.
- Objectives must remain within the project's defined scope.

1.4 Research questions

This section outlines the key questions the project will answer. These questions guide the research and project focus, directly aligning with the specific objectives and helping to frame the study's direction.

1.5 Scope of the project

1.5.1 Content Scope

Defines the specific topics or areas the project will cover and limitation.

1.5.2 Time Scope

Indicates the timeframe within which the project will be conducted or analyzed.

1.5.3 Geographical Scope

Specifies the location or region where the project's study or data collection will be focused.

1.6 Significance of the project

1.6.1 Research interest

Highlights the project's relevance to the researcher's personal goals or professional growth.

1.6.2 Institutional interest

Explains how the project will benefit the institution or contribute to institutional goals.

1.6.3 Future researchers

Emphasizes the project's value in providing a foundation or reference for future academic or industry-related research.

1.8 Organization of the Project

Provides an outline of the report's structure, briefly summarizing the contents of each chapter. This section helps readers navigate the document and understand how the project is organized, from introduction to conclusion.

CHAPTER 2: THEORETICAL CONCEPTS AND LITERATURE REVIEW

2.0 Introduction

Provides a brief overview of the chapter, highlighting the purpose of the literature review and setting the context for the research by briefly outlining the background, relevance, and key concepts. It should give readers a clear sense of what to expect in the following sections.

2.1 Theoretical concepts

This section defines and explains the main theories and concepts that form the foundation of the study. It is essential for linking the theoretical foundation with the objectives of the research.

2.2 literature review

In the literature review, include a minimum of four relevant studies that directly relate to your project's topic. Each study should be carefully selected to cover various aspects of the research theme and contribute valuable insights.

Here's what to include for each study:

1. **Study Summary** - Briefly describe the purpose, methodology, and key findings.
2. **Relevance** - Explain how the study aligns with your project's focus or addresses similar issues.
3. **Comparison** - Compare findings, identifying similarities or differences with other studies.
4. **Gaps and Limitations** - Point out any gaps or limitations and how your study aims to expand on or address these areas.

Each study should be referenced in APA format to ensure proper academic integrity and readability.

2.3 Critical review

Critically analyzes and evaluates the strengths and weaknesses of the literature covered. This section highlights any inconsistencies, gaps, or limitations in previous studies, emphasizing how the current project intends to address these gaps.

CHAPTER 3: METHODOLOGY AND ANALYSIS OF THE SYSTEM

3.1 Introduction

Provide an overview of the methodology chapter, summarizing the purpose and approach used in developing the project. Briefly describe the methods of data collection, tools, and techniques for the system's development.

3.2 Data collection techniques

Explain the different methods used to gather data for the project. Each technique should be described with a focus on its relevance and contribution to understanding the project requirements. *(The following are the techniques stated, describe what you applied only)*

3.2.1 Interview

Describe the purpose of conducting interviews, the target respondents, and the types of questions asked to obtain direct information about the project needs. Describe how you will design interview questions, conduct the interviews, and analyze responses to inform system requirements.

3.2.2 Documentation

Detail the use of existing documents, reports, or records to gain insights into the current system's workflow and historical data. State how this documentation will be reviewed to identify gaps, compliance standards, or requirements that should be incorporated into the proposed system.

3.2.3 Observation

Mention that direct observation will be carried out to study current processes, user behaviors, or challenges faced within the existing system. Describe how observations will be recorded and analyzed to gain a real-time understanding of workflows and areas needing improvement.

3.2.4 Internet research

Indicate that online research will be used to explore similar systems, best practices, or emerging technologies relevant to the project. Explain how findings from credible sources will inform the design and feature set of the new system.

3.2.5 Group discussions

Describe how structured group discussions with team members or stakeholders will be held to brainstorm ideas, validate assumptions, and refine the scope. State how you will document and incorporate feedback from these discussions into the project design.

3.3 Software engineering methodology

Specify the software development methodology (e.g., Agile, Waterfall) chosen for the project, explaining why it was selected and how it guides the project's stages from design to implementation.

3.4 Existing system overview

Provide an overview of the current system being used, if applicable, to identify gaps and areas for improvement.

3.4.1 Introduction

Introduce the existing system, summarizing its main purpose, features, and users.

3.4.2 Current system workflow

Describe the workflow of the current system step-by-step to show how processes are currently managed.

3.4.3 Limitations and challenges

List and explain the limitations and issues in the current system that highlight the need for an improved solution.

3.5 Proposed system description and improvements

Describe the new system, emphasizing the improvements and new features it brings. Highlight how it addresses the limitations of the existing system.

3.6 System requirements

Detail the requirements for setting up and running the proposed system, divided into hardware and software needs.

3.6.1 Hardware Requirements

List the hardware specifications required to operate the system, such as processor type, memory, storage, and other essential hardware components.

3.6.2 Software Requirements

Specify the software needed, including operating systems, programming environments, and additional software dependencies.

3.7 Functional Requirements

Describe the essential functions the proposed system will perform, which define what the system should be able to do for users.

3.8 Non-functional Requirements

Outline the proposed system's quality attributes, such as performance, usability, security, and scalability, which determine how the system performs its functions.

3.9 Use Case Diagram

Present a use case diagram to illustrate the interactions between users (actors) and the system, showing different user roles and their access to various system features.

3.10 Data Flow Diagram

Provide a Data Flow Diagram to represent how data moves through the system, showing input, processing, and output stages.

3.10.1 Level 0 (Context Diagram)

Create a high-level context diagram that shows the entire system as a single process and the interactions it has with external entities.

3.10.2 DFD Level 1

Expand the Level 0 diagram by breaking down the main process into sub-processes to show more detailed data flows within the system.

3.11 Entity relationship diagram

Include an Entity Relationship Diagram (ERD) to illustrate the database structure, showing entities, attributes, and relationships between different data elements in the system.

PART VI: CONCLUSION

In conclusion, the FYP proposal guidelines provide a clear framework for selecting, developing, and presenting projects. By following these criteria, students will ensure their topics are relevant, feasible, and aligned with departmental goals. Adhering to formatting and referencing requirements will maintain academic integrity and clarity, helping students produce high-quality, well-organized projects.